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Review

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Evolution in space: experimental pathways to understanding life's adaptability beyond Earth

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Understanding how life might evolve beyond Earth is a long-standing question in astrobiology. While significant advances have expanded the search for potentially habitable exoplanets, the biological and evolutionary processes that life may undergo in extraterrestrial environments remain largely speculative. Despite early recognition of its importance, experimental studies on evolution in space and analogue environments (AEs)—particularly involving multicellular organisms—have been limited. This gap is due in part to the difficulty of replicating complex space and other planets' conditions, such as microgravity, high salinity, pressure and/or radiation, on Earth. Recent spaceflight and AE experiments, however, have begun to yield critical empirical data. In this review, we integrate theoretical and experimental data from Earth and space studies to explore the evolutionary trajectories life might take beyond Earth. We interpret current knowledge from ground-based, spaceflight and AE biological studies across a wide range of organisms, and emphasize the need for multigenerational studies for accurately predicting and detecting diverse forms of life beyond Earth.

1. Introduction

How does life evolve on other planets? Can we predict the phenotype of alien life based on our knowledge of life as we know it? These questions are tied to the search for new habitable worlds that could harbour life and by extension astrobiology. In the last decade, while the advances on the astronomical and space engineering fronts have been fruitful through various impressive missions targeting ocean worlds that could harbour life [1,2], the biology and the evolution of life itself on these planets have remained vastly speculative. Aside from studies targeting extremophile organisms [3], we still have a limited understanding of how life—in particular complex life—can evolve in space and other environments [4,5]. One of the main reasons for this lack of understanding is the difficulty of reproducing space conditions on Earth. Factors such as altered gravity, radiation, the presence or absence of magnetic field, salinity and—most importantly—their combined effects are challenging to simulate in terrestrial laboratories. This gap of knowledge impairs not only our understanding of the evolution of life elsewhere, but also space travel as the long-term effects on settlements on life, including humans, are currently unknown [4].

One way to circumvent this is to mimic the evolution of life as we know it in space or other planets' analogue environments (AEs) in multiple generations. These approaches can allow us to redefine the limits of life (or how these boundaries evolve, see below) and can refine predictions regarding the evolution of life in extraterrestrial environments. This idea has been discussed

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in theoretical biology, space biology, evolutionary biology and more recently astrobiology, but has never been developed at a large scale [3,6,7]. This gap limits our knowledge of the trajectories by which life, including humans, might evolve in space and beyond. In addition, while some theoretical studies have investigated this last aspect and provided interesting theoretical results [5,8–10], they often solely consider random mutation rate and simple models that do not recapitulate and take into account the complexity of the organisms or the environment.

In recent decades, experiments conducted aboard space stations and in AEs have begun to yield critical empirical insights about the evolution of life in extraterrestrial environments. These results offer critical insights into the potential evolutionary trajectories of life in space and other planets, allowing us to refine our understanding of extraterrestrial life and the biological challenges of human health and survival during space travel. Here, we begin by exploring how evolutionary trajectories might unfold in space, considering recent work that incorporates both classical evolutionary forces and the increasingly recognized roles of developmental constraints and plasticity. We interpret current knowledge from ground-based and spaceflight biological studies across a wide range of model organisms, emphasizing that a deeper integration of biology into astrobiology is essential for accurately predicting and detecting diverse forms of life beyond Earth.

2. Why studying evolution in space and extraterrestrial environments

Species evolve and diversify in continuous interaction with their environment. They get resources and energy from it, are modified by it and can, in turn, modify this environment. As environments change over time, the equilibrium between species and their surroundings shifts, influencing the evolution of phenotypes along specific paths (figure 1A; see box 1 for definitions). Space and habitable planets present environments that can differ significantly—sometimes radically—from those of Earth. While often extreme, conditions such as varying gravity, pressure, radiation, temperature extremes or elevated salinity can act as selective pressures, influencing the evolutionary paths of life at all levels. In these contexts, the phenotypes with the highest fitness (i.e. those most likely to survive and reproduce) will thus differ from those of Earth, leading to new adaptive peaks in space (figure 1B, in which adaptive peaks and phenotype distribution are shifted compared to Earth) or on other planets (figure 1C, in which adaptive peaks and phenotype distribution are shifted compared to Earth and space). Many examples support this idea, such as the evolution of extremophiles in environments similar to what is found on other planets [3], the evolution of frogs and other organisms in Chernobyl [14,15], the evolution of organisms in space [16] or in closely monitored conditions such as clean rooms or aboard the ISS [17,18]. Understanding these environmental effects, and by which mechanisms they can influence evolution, is essential for predicting how life might evolve in space or on other planets.

Box 1. Definitions

Adaptive landscape/peak: Representation of the relationship between the mean fitness of a population and the allele frequencies at one or more loci that affect fitness. Populations with allele frequencies that maximize the mean fitness are represented as peaks (adapted from [11]).
Biological constraints (physical, biological and developmental): A restriction to produce a large variety of phenotypes due to the inherent properties of a given system (e.g. physical, cellular, developmental). Constraints generally tend to direct the evolution of phenotype towards preferential paths of least resistance.
Canalization: The evolution of internal factors during the development of organisms that reduce the effect of perturbing environmental and genetic influences, thereby constraining variation and consistently producing a particular phenotype (adapted from [11]).
Contingency: When rare historical events and chance occurrences result in unique and predictable evolutionary pathways.
Convergent evolution: Evolution of similar phenotypes in different lineages independently, in general following similar environmental pressures, usually from different genes or developmental processes.
Determinism: When evolution produces recurrent outcomes under similar selective pressures resulting in predictable pathways.
Divergent evolution: Evolution of populations sharing a common ancestor, accumulating genetic and phenotypic differences after adapting to distinct environments or niches.
Epigenetic/genomic: Heritable changes in gene activity that occur without changes to the DNA sequence.
Genetic drift: Random change of frequency of two or more alleles or genotype within a population (adapted from [11]).
Mutation: An alteration/change in the genome that is not due to reciprocal recombination.
Natural selection: The differential survival and/or reproduction of classes of entities that differ in one or more characteristics. This difference cannot be due to chance and must have the potential consequence of altering the proportions of different entities. The entities may be alleles, genotypes, populations of species (adapted from [11]).
Plasticity: Phenotypic plasticity is the ability of a genotype to produce different phenotypes in response to distinct environmental conditions. Phenotypic plasticity arises from the developmental process, which integrates genetic and environmental information to generate a phenotype [12].
Plasticity-led evolution: Plasticity-led evolution occurs when a change in the environment triggers a change in phenotype via phenotypic plasticity, and this pre-existing plasticity is subsequently refined by selection into an adaptive phenotype [13].

Over the past few decades, a growing body of research across multiple organisms has transformed our understanding of the mechanisms by which environmental conditions influence the emergence of new phenotypes. These studies, both experimental

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and theoretical, revealed that evolution acts not only at the genomic but also developmental and phenotypic levels (reviewed in [19–21]). These considerations have important implications for evolution in space and extraterrestrial environments, as well as for designing experiments to study them. Classical evolutionary forces like mutation, natural selection and genetic drift generate genetic variation, with natural selection favouring traits that enhance survival in specific environments (box 1). In addition, other evolutionary driving forces, such as physical, biological and developmental constraints as well as plasticity, filter or alter the expression of genotypes and the distribution of phenotypes in each environment (see figure 1A–C for theoretical examples of how advantageous phenotypes change depending on the environment). Mathematical, chemical, physical and biological laws limit the emergence of certain life forms. For example, temperature can influence organismal sex, while developmental constraints, such as embryogenesis, canalize phenotypic evolution towards specific trajectories [20–25]. In novel or changing environments, particularly those with radically different physical and chemical conditions, these paths can shift [21], redirecting evolutionary trajectories towards adaptive peaks, sometimes convergently [19,26]. In extraterrestrial environments, both external and internal constraints may modify these adaptive peaks, directing the evolution of phenotypes by changing the mutation rate, species survival, canalizing the shape of organisms or modifying development and species survival (figure 1). Understanding how these shifts shape phenotypic trajectories, not only at the genetic but also at various levels of organismal development, is essential for refining predictions of how terrestrial life might adapt to alien environments. Given the convergent nature of evolution [27], with species evolving similar traits in similar environments, experimental work in space or AEs is crucial for testing these hypotheses.

Up to now, and despite numerous attempts and suggestions [6,28–30], experimental evolution in analogue extraterrestrial environments have remained sparse, in particular for multicellular organisms. One reason for this delay is the perception that biological evolution is inherently slow, with significant effects only emerging over extended periods, making short-term experiments seem less significant. However, while evolutionary forces shape organisms over extended periods of time, recent studies have revealed that phenotypes can evolve rapidly under anthropic pressures and rapidly changing environments—sometimes within one or a few generations as exemplified by work on the evolution of animals under poaching pressure or in radiated zones [14,15,31,32]. Another important aspect of this debate is that evolution is inherently difficult to predict, being the result of both determinism and contingency [20,26,30]. Constraints and convergences often shape evolution by limiting the range of possible phenotypes, as shown by many cases of parallel evolution. Yet, genetic differences among populations can lead to divergent outcomes, even under similar environmental pressures. Evolution thus reflects a balance between determinism, where similar conditions yield similar results, and contingency, where unique histories lead to different paths [26,33].

Up to now, the study of evolution of complex organisms in space mainly relies on computational simulations [5,34]. Although these models provide valuable insights into the potential consequences of mutations in humans, they cannot fully recapitulate the complexity and emergent properties of evolving biological systems. For all these reasons, there is a need for experimental evolution in space and AEs to be able to understand how life adapts to space, and evolves in space and extraterrestrial environments.

3. Experimental evolution and change in evolutionary trajectories

To anticipate how organisms may adapt to challenging conditions of extraterrestrial environments, including space, we need to investigate how evolutionary processes unfold across multiple generations. This goal can be explored through experimental evolution, a long-established approach in which populations are maintained under controlled experimental conditions to observe adaptation in real time [33,35]. In such experiments, selection acts on all traits affecting fitness under the imposed regimes, while other evolutionary forces such as mutation, genetic drift and gene flow operate simultaneously, as they do in nature. Such designs have been used to examine adaptation to specific stressors, including temperature shifts, nutrient limitation or radiation, making them particularly relevant to astrobiology. Beyond measuring phenotypic and genomic responses to controlled environmental pressures, experimental evolution also provides a powerful tool to explore trade-offs, where traits beneficial in one environment are costly in another, to identify interdependencies among adaptations, and to quantify the effects of beneficial mutations on evolutionary trajectories [35–37]. By extending these approaches to conditions that approximate spaceflight or planetary environments, we can begin to predict how life might evolve beyond Earth.

3.1. Long-term microbial evolution

Long-term experimental evolution studies can help to connect microevolutionary patterns within populations and species to macroevolutionary processes, such as the emergence of new species, when microbes are exposed to new environments [33]. A classic example is the long-term evolution experiment, in which 12 populations of *Escherichia coli* were grown in a low-glucose environment to study adaptation, historical contingency and patterns of parallel and divergent evolution [37,38]. The experiment revealed parallel changes on morphological and genomic levels, and interesting specializations, including numerous beneficial mutations. One of the lines eventually developed the ability to metabolize citrate in addition to glucose as the main source of energy, which is not typical for *E. coli* [38]. This adaptation required earlier potentiating mutations suggesting that historical contingency shapes evolutionary trajectories. In challenging environments, evolutionary outcomes often depend on the order in which mutations arise, making them potentially unrepeatable. This helps to explain why the ability to metabolize citrate evolved only after more than 30 000 generations [38]. Currently, the experiment has passed 50 000 generations, with fitness continuing to increase, suggesting that the rate of adaptation slows but did not stop over time. Experimental evolution has also been used to generate the most radiation-resistant *E. coli* populations to date, revealing insights

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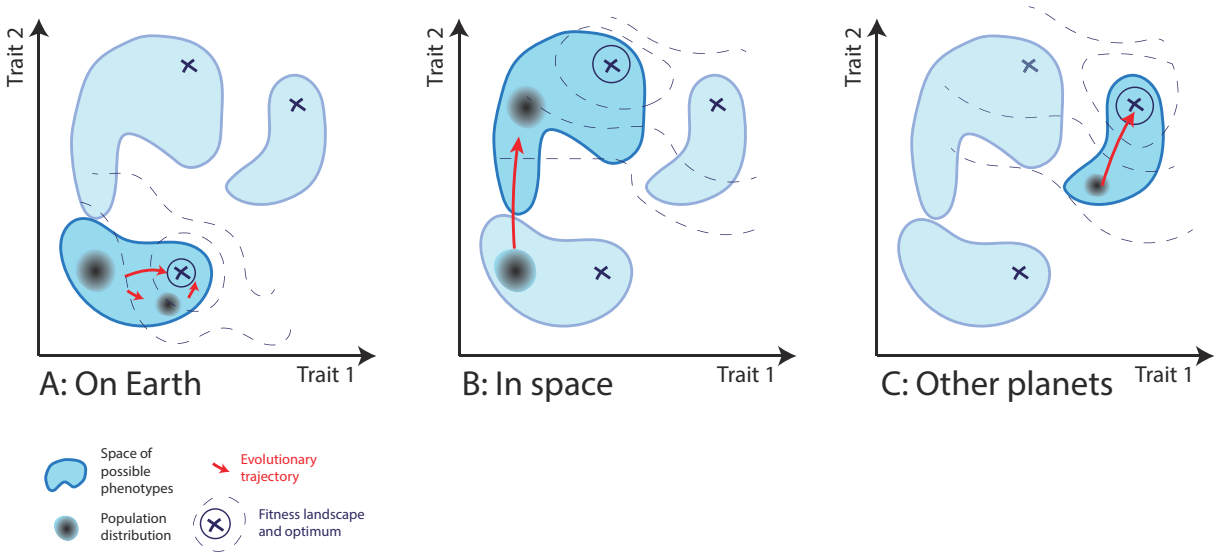


Figure 1. The space of possible phenotypes, or morphospace, illustrated here for two traits, is constrained by biological limits. This produces common phenotypes that are easily reachable and rare phenotypes that are less accessible, depending on the environment and its associated fitness landscape. On Earth (A), the evolutionary optimum can be reached from two populations following different trajectories. In space (B), shifts in selective pressures, physical conditions and developmental constraints may allow species to explore new regions of morphospace. On other planets (C), species could evolve phenotypes that are inaccessible under Earth conditions.

about how microbes could adapt to high doses of radiation [39]. After 100 cycles of selection under high-dose irradiation, replicate populations evolved tolerance to doses of about 3000 Gy. These populations exhibited several specific traits, including long-lived subpopulations and the accumulation of porphyrin compounds that produced brown-pigmented colonies. Genomic and physiological changes included variants of proteins involved in ATP synthesis, iron–sulfur cluster assembly and cadaverine synthesis, all contributing to enhanced radiation resistance. The evolved populations also displayed improved protection of cellular proteins from oxidative damage, resembling the naturally radioresistant bacterium *Deinococcus radiodurans* [39,40]. Remarkably, the experiment is ongoing, with no upper limit to radiation resistance detected so far. These experiments show that experimental evolution can reveal how life adapts to novel and extreme conditions, providing key insights into how organisms might evolve, diversify and persist in extraterrestrial environments.

3.2. Multicellular experimental evolution

At the multicellular level, a classical long-term study of 600 generations investigated the trade-off between postponed reproduction and ageing in *Drosophila melanogaster*, compared to early-breeding populations. The results initially confirmed that delayed reproduction was associated with extended lifespan and slower ageing rates. However, over time, these correlated evolutionary changes began to reverse, and age-related mortality eventually plateaued, suggesting limits to the plasticity of lifespan extension [36]. More examples include laboratory evolution experiments in nematodes, where populations were subjected to fluctuating anoxia–normoxia cycles, elevated salinity and high-temperature stress. These studies demonstrated the capacity of nematodes to adapt to a range of harsh environmental conditions over multiple generations, providing insights into the genetic and physiological mechanisms underlying stress resistance and environmental tolerance [41]. Yet another compelling example is the ongoing Multicellularity Long-Term Evolution Experiment (MuLTEE), which investigates the mechanisms and dynamics of the evolutionary transition to multicellularity. In this study, replicate populations of 'snowflake' yeast (*Saccharomyces cerevisiae*) mutants, which naturally form fractally branching multicellular clusters, are grown daily while selected for increased cluster size. Throughout more than 3000 generations, these initially small and fragile clusters evolved to become tens of thousands times larger and as mechanically tough as wood [42]. More recently, another MuLTEE experiment from the same group has revealed that a founder whole-genome duplication event can provide immediate adaptive advantages, persist under selection and facilitate long-term innovation by expanding heritable genetic variation. Together, these experiments have revealed how multicellular groups can progressively acquire novel, highly heritable multicellular traits via genetic and epigenetic mechanisms, providing a powerful model for studying the origins and integration of multicellular life.

3.3. Applying these concepts to extraterrestrial environments

Experimental evolution has proven to be a powerful tool for studying adaptation to defined selection pressures, and applying this approach to space environments could be equally valuable. Long-term experiments under spaceflight or other planetary environments could reveal unexpected evolutionary innovations beyond the limits of short-term experiments. Spaceflight or extraterrestrial conditions can be applied as an environmental or selection factor in experimental evolution approaches—either separately or simultaneously—being representative of the spaceflight environment. The environments of other planets could be simulated either directly or through gradual transitions, enabling the generation of life forms adapted to such conditions.

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4. Evolutionary trajectories are modified in space and analogous environments

Over the past three decades, a wide variety of organisms—including bacteria, plants, amphibians and vertebrates—have been sent to space or studied in AEs (figure 2A,C represents a summary of clades that have been sent to space the observed phenotypes). As previously mentioned, space or terrestrial environments influence the evolution of organisms at all levels: genomic (their DNA) through mutation and epigenomic changes, developmental by altering their morphogenesis, physiology and phenotype through fitness and survival and morphological by influencing fitness and survival (figure 2B exemplifies the effect of the environment at all these levels). Because of this, various species have been sent to space and studied at various stages of their life cycle (from the egg to the adult; figure 2A,C). Although comparing the results of these studies is challenging due to the sporadic nature of missions and varying experimental conditions (table 1), they offer valuable insights into the potential evolution of life in extraterrestrial environments and the impact of space and other planets—conditions on organisms' reproduction, development, phenotype and eventually survival.

4.1. The evolution of prokaryotes in space provides insights about adaptation

As the life cycle of prokaryotes is short, they are among the few groups (but see [33]) of organisms for which we can explore which evolutionary trajectories are canalized in space and extraterrestrial conditions. Experiments performed on *E. coli* for over 1000 generations under low shear simulated microgravity have revealed the existence of adaptive mutations, demonstrating that, in space, bacteria may evolve along different evolutionary trajectories from those of their Earth counterparts [16]. While the mechanisms driving these adaptations are poorly studied, reports show that microorganisms in space undergo global metabolic and gene expression changes at both transcriptional and translational levels [66] including altered DNA repair mechanisms, oxidative stress responses and membrane composition, indicating metabolic network reprogramming [66] (figure 2A). Simulated microgravity, both short- and long-term, has also been linked to changes in growth rate, biofilm formation, gene expression and key physiological processes, including stress response, membrane transport, metal ion transport, adhesion and metabolism of carbohydrates and lipids [67]. Moreover, the extreme environment of the ISS puts unintentional selective pressure on microorganisms introduced during its 20 years of service at LEO. It has been shown that such pressure leads to the development of new features that help microorganisms adapt to challenging conditions, not found in terrestrial relatives [17,68]. For example, a study summarizing multiple in-flight and simulated spaceflight experiments on *E. coli*, *Bacillus subtilis*, *Salmonella enterica*, *Staphylococcus aureus*, *Pseudomonas aeruginosa* and *Ralstonia pickettii* found that spaceflight conditions significantly alter bacterial morphology, growth, metabolism, gene transfer and antibiotic resistance [3] (figure 2A). Some strains exhibited enhanced biofilm formation, altered virulence and antagonistic behaviour, underscoring the adaptive plasticity of microbes in space environments and the need for careful microbiome management on long-duration missions [3]. Recently, two studies found that the harsh environments of the ISS led to the emergence of new species. A functional annotation of the genomes of five newly identified species of Gram-positive bacteria was generated, four of which are non-spore-forming and one spore-forming, isolated from the ISS [68]. Comparative genomic and proteomic analyses revealed shared features across five species as well as traits unique to ISS microorganisms, including adaptations to hypoosmotic stress, enhanced DNA repair, activation of mobile genetic elements and pathogenicity-associated functions such as peptide synthesis and ATP-dependent transporters. These findings suggest that microbes exposed to spaceflight conditions may acquire increased antibiotic resistance and pathogenic potential [68] (figure 2A). These findings are reinforced by a recent large-scale survey of the ISS, which produced the most extensive dataset so far on a space habitat's microbial and chemical composition, based on 803 surface samples collected from nine modules [69]. The study revealed clear module-specific microbial and chemical signatures shaped by human activity and function, with habitation, hygiene and laboratory modules hosting distinct communities. When compared with thousands of terrestrial environments, the ISS microbiome showed a sharp reduction in diversity and a dominance of human-associated taxa, placing it at the extreme end of the built-environment gradient. Comparative genomics further indicated that several ISS bacterial strains are phylogenetically distinct from their Earth counterparts and harbour unique antimicrobial resistance and metal-tolerance genes, suggesting early stages of divergence under prolonged space habitation. Together, these results reveal that prokaryotes undergo change and selection in space, and even evolved new species and lineages that are specific to the ISS (see figure 2A for a summary of these results).

4.2. Alteration of the developmental trajectories of multicellular organisms in space

Multigenerational studies in multicellular organisms have been far more limited than in prokaryotes. Although a wide range of species have been tested in orbit including *Arabidopsis thaliana*, various crops, flies, nematodes, zebrafish, frogs, geckos, newts, killifish, quails, mice and rats (figure 2C, table 1), very few of these experiments have extended across multiple generations. Until recently, technological constraints also prevented comprehensive genomic monitoring of these organisms, making it difficult to trace how evolutionary changes arise under space conditions. As a result, the study of multicellular evolution in space has progressed more slowly and remains fragmentary compared with microbial systems. Nevertheless, ISS experiments have provided valuable insights, identifying developmental stages that are most vulnerable to perturbation, suggesting potential adaptive trajectories and revealing mechanisms by which organisms may adjust to extraterrestrial environments (figure 2B). The main phenotypes and experiments for the organisms mentioned in our present review are detailed in table 1. Other, more exhaustive databases can be found in excellent reviews of space biology [43–45].

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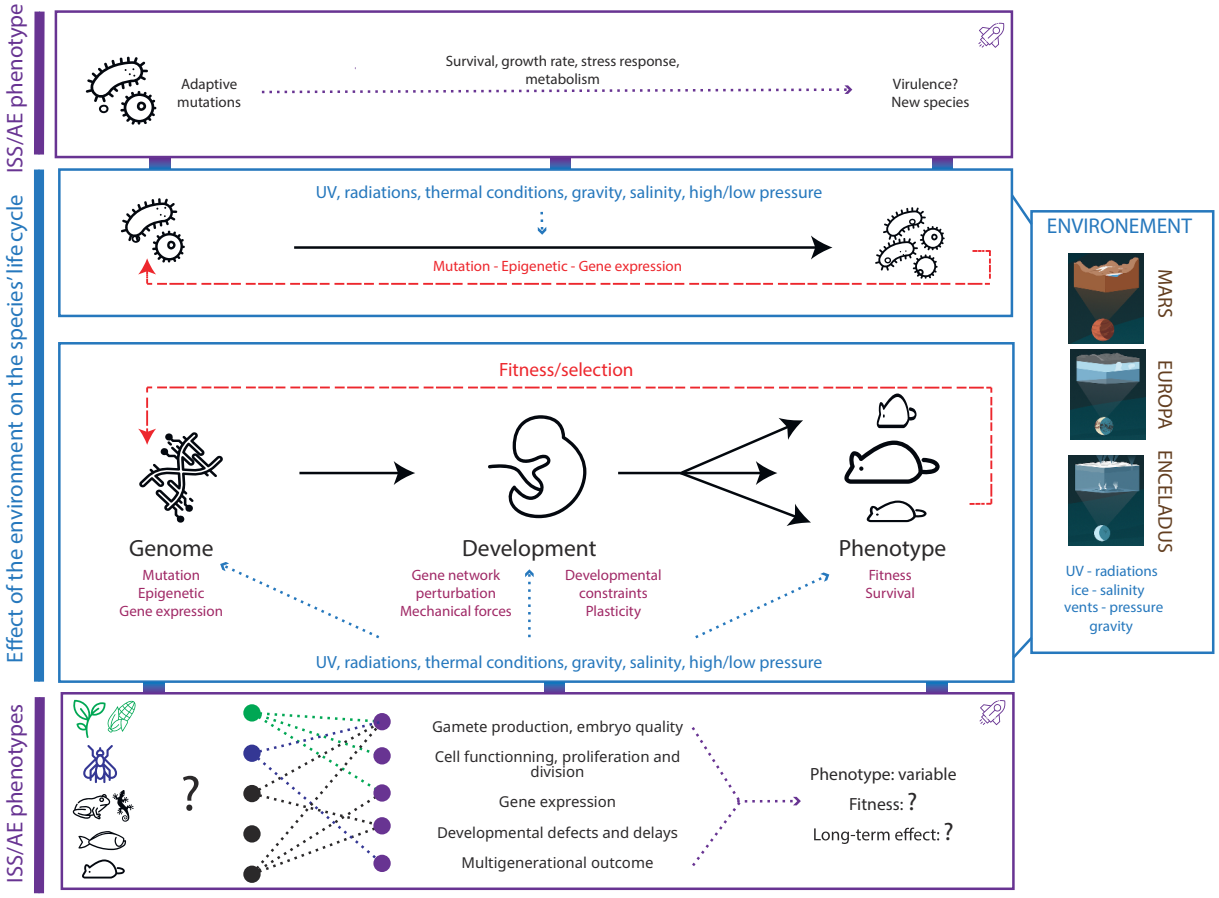


Figure 2. Effect of space and extraterrestrial environment on species' life cycle. Upper panel: ISS and AE evidence on prokaryotes. Middle panel: Interaction between space and extraterrestrial environments and the processes that contribute to organismal evolution. Lower panel: ISS and AE evidence on multicellular organisms.

4.2.1. Spaceflight phenotypes across species

In plants, the first characterized space phenotypes came from early experiments in lettuce and lentils, where seeds displayed atypical morphologies and impaired gravitropism when germinating [46,47]. Subsequent ISS experiments with *A. thaliana* established that plants can complete a full life cycle, seed-to-seed, in orbit [49], but the quality of embryos and seeds is frequently compromised [48,50,70]. Crops such as *Brassica rapa* showed impaired pollination mechanics and reduced fecundity [71]. Morphological alterations extend to root growth, leaf expansion and flowering time, reflecting broad physiological sensitivity to gravity absence. These results reveal that plants, while capable of maintaining a complete life cycle, exhibit alteration of their development and phenotype when developing in space. In flies, decades of studies revealed that spaceflight significantly alters gene expression and physiology, in particular immune response, neurobehaviour, the cardiovascular system and, to a lesser extent, development (reviewed in [45]). In amphibians, a large number of experiments have been performed on multiple species of frogs and newts at various developmental stages. While sometimes difficult to compare, these studies reveal that while frogs ovulate successfully in orbit, mid-developmental stages (gastrulation, neurulation) often exhibit mild defects [58,59]. In contrast, *Xenopus laevis* tadpoles grown in orbit exhibit severe abnormalities, including enlarged craniofacial structures, eyes and neural defects back on Earth [60] (reviewed in [44]). In newts, while early development appears normal, neural defects have also been observed, with modification of the neural tube similar to what can be seen in frogs but more severe [61]. Despite these abnormalities, many amphibians recover upon return to Earth, pointing to a robustness of developmental systems though it remains unclear how complete orbital development would shape adult phenotypes. In birds, quail embryos develop but with delays, nutrient deficiencies and vestibular abnormalities, and hatchlings often show behavioural anomalies [44;63, and references therein]. In fishes, fertilization and hatching occur normally, though phenotypes vary across species or lines [44,62]. In mammals, rats exposed during gestation exhibit vestibular and inner-ear malformations, and mice from shuttle missions showed disrupted ovarian and uterine function, impairing fertility [64]. Together, these cases reveal a spectrum of space-induced phenotypes, ranging from subtle shifts in developmental timing to anatomical defects, some reversible on Earth and others with lasting consequences.

4.2.2. Developmental robustness, plasticity, and their responses to spaceflight

Inspection of spaceflight phenotypes reveals that development is not uniformly affected by environmental perturbations such as microgravity and radiation. Instead, certain developmental stages are markedly more sensitive than others [60]. Core, often early, developmental stages exhibit strong canalization and resist perturbation, whereas others, such as neurulation, are more variable and susceptible to disruption. Perturbations at these sensitive stages may cause phenotypic divergence and potentially

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Table 1. Spaceflight phenotypes of plants and animals discussed in the present study. Summary of developmental stages tested and phenotypic outcomes observed in plants and animals exposed to spaceflight or simulated microgravity. References correspond to key experimental reports discussed in the text. For more comprehensive reviews, see [43,44,45].

clade	organisms	developmental stages	reported phenotypes in space/microgravity	references
plants	lettuce, lentils	germination	atypical seed morphology, impaired gravitropism	[46,47]
	Arabidopsis thaliana	full seed-to-seed cycle	life cycle completion, but reduced embryo/seed quality, impaired fertilization, altered flowering, root growth and leaf expansion; cell wall biosynthesis, auxin transport and hormone signalling disrupted	[48,49]
	Brassica rapar	reproduction	impaired pollination mechanics, reduced fecundy	[50]
	other crops (wheat, pea, rice)	seed-to-seed cycle	altered branching, silique orientation, smaller starch-retaining seeds, variable flowering times, inconsistent root growth	[50–53]
ecdyzozoa	fruit fly (*Drosophila melanogaster*)	egg to adult; up to three generations	development sustained, but altered gene expression (morphogenesis, cuticle, cytoskeleton); immune, neurobehavioural, cardiovascular changes	[45,54]
	nematodes (*Caenorhabditis elegans*)	up to 12 generations	morphologically normal; altered cytoskeletal and cuticule organization, miRNA expression, muscle development; conserved stress responses	[55–57]
amphibians	frogs (*Xenopus laevis*)	fertilization to tadpole	ovulation successful, but defects at gastrulation/neurulation; tadpoles hatched on Earth show craniofacial and neural abnormalities	[58–60]
	newts (*Pleurodeles waltl*)	embryogenesis	early stages normal; later neural tube defects, more severe than frogs	[61]
fishes	zebrafish, killifish, others	fertilization to hatching	fertilization and hatching normal; phenotypes variable by species/line	[44,62]
birds	quails	embryogenesis to hatching	developmental delays, nutrient deficiencies, vestibular and behavioural abnormalities	[44,63]
mammals	rats	gestation	vestibular and inner-ear malformations	[64]
	mice	sperm maturation	disrupted ovarian/uterine function; impaired fertility; sperm matured in space causes offspring gene expression and epigenetic changes (DNA damage, development)	[64,65]; M.M., unpublished

fuel adaptive evolutionary change [72] (table 1; see [73] for an example). In plants, fertilization, gametogenesis and early embryo development are particularly sensitive, with pollen tube growth and ovule fertilization often impaired [50,70]. Similarly, meristem development and cell cycle progression are frequently affected. Mars- and Moon-gravity simulations further show that partial gravity modifies the cell cycle of *A. thaliana*, suggesting that key developmental pathways are tightly constrained by gravity [34,74,75]. In amphibians and birds, neurulation is particularly vulnerable, with microgravity exposure during these stages causing patterning defects [59] (but see [44]). In mammals, preimplantation and early gestation are most affected: microgravity slows cleavage and blastocyst formation, reducing implantation success [76]. Rat studies show, however, that once implantation occurs, later gestation is comparatively resilient [64]. Despite such perturbations, many organisms are able to complete their life cycles once back on Earth, underscoring the robustness and buffering capacity of developmental systems at earlier and later stages. In plants, microgravity strongly reprogrammes auxin transport and hormonal signalling, altering cell proliferation and differentiation, but still leading to a viable adult morphology [34,74,77]. Similarly, environmental cues such as light can restore balance: red light exposure rescues root proliferation and auxin flux under simulated microgravity [77]. In flies, improved experimental hardware allows development from egg to adult in microgravity and even hypergravity, showing that life cycles can be maintained despite extensive molecular changes affecting metabolism, sperm development and pupal cuticle [54] (Saclier *et al.*, unpublished data). In birds, quail hatchlings often compensate behaviourally for neural developmental delays [44]. Such buffering enables reproduction and may itself evolve under altered environments, yet hidden costs may emerge in fitness-related traits or across generations. These results suggest that certain phases of development are less constrained and therefore freer to evolve in response to environmental changes. They also point to developmental responses that may begin as plastic but later become stabilized by natural selection, as observed in other species on Earth [13].

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4.2.3. Convergent spaceflight phenotypes

Finally, spaceflight produces some common phenotypes across phylogenetically distant species, suggesting that space phenotypes can be anticipated, at least in similar environments (reviewed in [43–45]). Several convergent outcomes are repeatedly observed across plants and animals: delays in development, disruption of reproduction, sensory and neural defects, disrupted mitochondrial function, oxidative stress and DNA damage responses, and remodelling of cytoskeletal and structural pathways. In plants, the meristem and gametogenesis are consistently affected. In animals, the vestibular system, early embryogenesis and reproductive physiology emerge more affected. These patterns suggest that the space environment consistently impacts a restricted set of developmental and physiological modules, generating comparable outcomes in plants and animals. This view is reinforced by recent studies comparing spaceflight-induced gene expression changes in *Caenorhabditis elegans* and humans, which identified conserved responses in a set of orthologous genes [78]. Although this study did not focus on multigenerational exposure, it supports the idea of an evolutionarily conserved consistent space-response across distant species. Using these patterns to anticipate which phenotypes are most likely to emerge under extraterrestrial conditions will be invaluable for astrobiology.

4.3. Multigenerational studies in multicellular organisms

Only a few studies have examined multigenerational effects in multicellular organisms, mainly in plants, nematodes and flies, with partial insights from mice. Up to now, only five species of plants, *A. thaliana*, wheat, pea, *Brassica rapa* L. and rice, have successfully completed seed-to-seed cycles in space [46,50–52]. These experiments revealed consistent morphological, reproductive and developmental changes, including altered branching and silique orientation, smaller seeds with retained starch, variable flowering times and inconsistent root growth patterns. At the cellular level, spaceflight disrupts key processes such as cell wall biosynthesis, membrane composition, cell cycle regulation, amyloplast distribution, cytoskeletal organization and gene expression, collectively indicating that microgravity acts as a strong stressor that remodels plant physiology from morphology to molecular pathways [53]. These findings indicate that plant responses to microgravity target conserved developmental and physiological modules repeatedly, which could constitute an adaptive evolutionary signature to the space environment. Similar conclusions can be reached for animals. As mentioned above, experiments spanning up to three generations show that flies can sustain development in microgravity, but exhibit significant gene-expression changes in morphogenesis, cuticle formation and cytoskeletal proteins, alterations that could ultimately trigger long-term evolution [54]. Nematodes that underwent up to 12 generations in space appeared morphologically normal [79], although recent analyses show that spaceflight triggers developmental stress responses, including cytoskeletal reorganization, altered miRNA expression and disrupted muscle development [55–57,80]. Notably, an integrative analysis of spaceflight omics datasets also identified alterations in cuticle formation, closely paralleling those observed in *Drosophila* [54,57], suggesting that conserved developmental pathways may shape organismal responses to space environments. Together, these findings corroborate the existence of space phenotypic signatures, which may later be stabilized through plasticity-led evolution. In mice, no intergenerational studies have been conducted properly, but analysing phenotypic changes in *in vitro* fertilization experiments using sperm that have been incubated for 35 days in space revealed changes in gene expression involved in DNA damage response and developmental processes as well as epigenetic changes in the offspring [65] (figure 1A; M.M., unpublished results). As illustrated in figure 3B, such changes could displace phenotypes towards new adaptive peaks in space.

Together, these studies indicate that microgravity and the space environment affect all living systems. Unicellular organisms adapt opportunistically in space, rapidly selecting traits that enhance survival, an evolutionary strategy that could be at play on other planets. In multicellular organisms, all phases of organismal development could be affected with variable outcomes, shaping both the phenotypes and evolutionary trajectories of affected species. Although partial recovery is sometimes observed while returning to Earth, the long-term effects on multiple generations—both phenotypic and evolutionary—remain unknown.

5. Analogue and space experiments are crucial to study the evolution in space and beyond

To investigate adaptation to space conditions, we emphasize the need for multigenerational studies across a wide range of unicellular and multicellular organisms, with regular sampling throughout development. Such studies should parallel approaches in experimental evolution, where controlled manipulation of organisms in laboratory or natural settings is used to track adaptive processes [3,7,29,35,43]. A variety of AEs can be used to study adaptation in extreme conditions similar to other planets [3]. These include terrestrial locations such as Antarctic dry valleys for their cold and arid conditions, the Atacama Desert for its extreme dryness and UV exposure, acidic hot springs and deep-sea hydrothermal vents simulating high-temperature environments, and hypersaline lakes reflecting the habitats expected on Mars or Europa [3]. These analogue sites provide critical insights into how life might originate, persist and adapt in extraterrestrial environments. They can also serve as testing grounds for evolutionary studies under extreme conditions and have been invaluable for understanding the evolution of early life on Earth and beyond.

More insights have been revealed as an outcome of the BIOMEX project, which has been a cornerstone in studying the survivability and adaptation of extremophiles and biomolecules in extraterrestrial environments, focusing particularly on Moon- and Mars-analogue conditions. The experiments conducted on the ISS and through ground-based simulations explored microbial, fungal and cyanobacterial responses to UV, vacuum and simulated Martian regolith, contributing crucial insights into planetary habitability and evolutionary adaptations [81]. The project established survival limits of diverse extremophiles and

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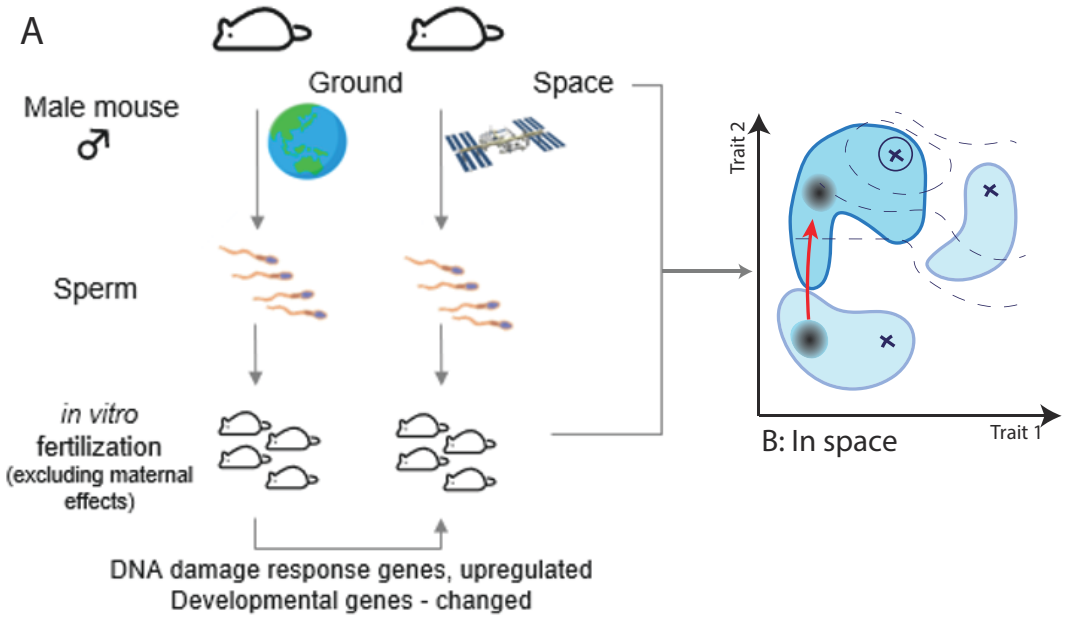


Figure 3. Transgenerational effects mediated by sperm exposed to space flight in mice. (A) Experimental design: the sperm of mice have been flown to space and then used to fertilize mice *in vitro* on Earth. The effect on the offspring will be compared with the appropriate ground control. (B) Potential shift in phenotype for the resulting litters that could result from mutational, epigenetic, plastic or developmental changes.

biomolecules under real space and Mars-like conditions aboard the ISS. Tested organisms, including fungi, cyanobacteria and lichens, demonstrated resilience through protective pigmentation, biofilm formation and DNA repair, highlighting evolutionary traits favouring survival in extraterrestrial environments [81]. For instance, cyanobacterium *Chroococcidiopsis* exposed to low Earth orbit conditions showed the capacity to tolerate vacuum, radiation and desiccation while maintaining viability and photosynthetic pigment integrity, with improved survival when mixed with minerals providing UV shielding [82]. The black fungus *Cryomyces antarcticus* demonstrated remarkable tolerance to radiation and temperature extremes, with minimal ultrastructural and DNA damage. Its resilience was largely attributed to thick melanized cell walls and metabolic quiescence, showing adaptations that could enable microbial survival in extreme extraterrestrial niches [83]. Finally, the resilience of the desert cyanobacterium *Chroococcidiopsis* to extreme UV radiation, vacuum and temperature fluctuations has been demonstrated, with mineral regolith analogues offering additional protection, suggesting ancient microbial lineages possess inherent adaptive traits that would be favoured in potential extraterrestrial environments or interplanetary transfer [84]. The project was a collaborative effort by an interdisciplinary consortium of 30 institutes from 12 countries, emphasizing that evolutionary studies on adaptation to space, due to their inherent complexity, require broad cooperation among researchers. These results reveal the importance of analogue experiments to refine the limit of life and adaptability on other planets and reinforce the idea that similar studies should be carried out for multicellular organisms, including plants and animals.

Studying experimental evolution in space or AEs should also be considered within the framework of planetary protection and bioethics. Investigating how organisms evolve beyond Earth carries inherent risks of both forward and backward contamination. These concerns are central to the Committee on Space Research (COSPAR) planetary protection guidelines, which emphasize minimizing biological release and preserving the scientific integrity of life-detection missions [85]. As emphasized by Cockell [86], ethical reflection on microbial life should not be limited to its instrumental or scientific value: even microbial forms, whether terrestrial or extraterrestrial, may warrant moral consideration as evolving entities with intrinsic value. Integrating evolutionary studies with these policies will be essential to ensure that research on life's adaptability in space proceeds responsibly, balancing scientific discovery with environmental safety and ethics beyond Earth.

6. Towards a unified framework to study evolution in space

The evidence from space and ground-based analogue experiments presented above highlights that understanding how complex life develops, reproduces and evolves in extraterrestrial environments requires a structured and integrative framework spanning multiple organisms across multiple generations. Such a framework will allow the field to move beyond descriptive accounts of physiological disruption and towards a mechanistic understanding of how the space environment shapes the evolution of life. Ultimately, it will allow us to predict the evolutionary trajectories of life in space and on other planets.

6.1. Establishing standardized procedures and reproducibility

A major limitation of earlier studies has been the heterogeneity of experimental conditions and reporting standards. The development of uniform procedures covering plant and animal housing, life support systems, sampling protocols, fixation methods and metadata collection is essential for comparability across missions and experiments [43,44]. The space environment combines altered gravity, ionizing radiation, vibration, CO_2 accumulation, humidity and temperature fluctuations. Robust

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monitoring of environmental and biological parameters is indispensable for interpreting experimental outcomes [44]. Establishing harmonized standards and linking datasets to open repositories, such as NASA GeneLab, will be essential to enable comparative studies across clades and species. Similarly, complementary ground-based analogues such as parabolic flights, clinostats and rotating wall vessels, along with countermeasure tests including shielding or controlled lighting regimes, should be standardized and systematically integrated into experimental designs. Equally important, experiments should be structured to address multiple biological questions (Sadier *et al.*, unpublished data).

6.2. Comparing life cycles and characterizing space response from genome to phenotype across species

To be able to compare life cycles between species the establishment of robust developmental benchmarks at all levels will be necessary. While protocols need to be taxon-specific as each species has unique housing, feeding, life-support requirements and reference milestones such as gametogenesis, fertilization, embryogenesis, hatching or birth, juvenile growth and reproduction can be used to align life cycles across taxa (such as the Carnegie stage system for vertebrates [87]). Multigenerational designs are another key step. Invertebrates with short life cycles can complete multiple generations within a single mission, providing insights into inheritance and evolutionary dynamics. For vertebrates with longer generation times, partial life cycle studies, such as gametogenesis or embryogenesis, could be combined across missions to build a composite view of the life cycle in space. Morphological measurements and atlases (X-ray computed tomography scan, histology), physiological readouts (growth rates, fertility, whole-organism and cellular metabolism), omics studies (genomics, transcriptomics, proteomics using cutting edge techniques) and behavioural assays should be integrated through modelling approaches and deposited in open repositories such as GeneLab. Such approaches would make it possible to identify conserved mechanisms, highlight lineage-specific vulnerabilities and predict how life adapts and evolves in space environments.

6.3. Application to planetary protection protocols and astrobiology missions

Multigenerational and experimental evolution experiments are particularly valuable for astrobiology, as they provide evidence of how life could look in extraterrestrial environments and help refine the signals that might be detected by instruments aboard space missions. Such experiments will also aid in detecting novel species that adapt and evolve in clean room environments, enabling their identification and decontamination prior to the launch of probes and rovers into space [18]. If such experiments are extended to space missions, they will also inform strategies for preventing back contamination of Earth with returned samples (see, for example, the strategies in place for Mars samples returns in [88] or the NASA Planetary Protection Handbook [89]). Indeed, while very unlikely, alien life could potentially be pathogeneous and/or compete with Earth life (see [88] and references therein). Finally, biosecurity issues must be addressed, including protocols for handling samples in BSL-3 or BSL-4 laboratories after their return to Earth or when novel or engineered organisms are involved, as these may pose risks to Earth's environment or human health [90].

Beyond planetary protection, studying evolution in space and extraterrestrial environments provides unique opportunities to study the limits of adaptability. These experiments can reveal which traits are most likely to evolve under non-terrestrial conditions, and generate models for how life might persist or fail in extraterrestrial environments. From an evolutionary biology perspective, they also shed light on how selection, drift, constraint and developmental plasticity operate under environments radically different from those of Earth. Integrating these insights into astrobiology offers an unprecedented opportunity to unify experimental space biology with the search for extraterrestrial life, thereby advancing a predictive evolutionary framework that connects the biology of Earth's organisms to the potential forms and trajectories of life beyond our planet.

7. Conclusion

Although the importance of studying evolution in space was recognized as early as the ISS era [29], major gaps remain, especially for multicellular organisms. Recent reviews emphasize the need for long-term, multigenerational studies under standardized protocols [4,7,43], and new initiatives and AEs now make such work increasingly feasible. Our synthesis shows that while evolutionary outcomes cannot be fully predicted, experimental studies across diverse taxa remain the most effective way to probe potential evolutionary trajectories in space. Unicellular organisms adapt opportunistically, rapidly acquiring traits that enhance survival, whereas multicellular species can reproduce in space but display inconsistent and species-specific responses. To anticipate how complex life might evolve beyond Earth, long-term, multigenerational experiments are indispensable. Such approaches refine our understanding of the limits of life and habitability, expand our knowledge of how life as we know it adapts to extraterrestrial environments, and reveal how selection, drift, constraint and plasticity operate under novel conditions. Integrating these insights into astrobiology will bridge evolutionary biology, space biology and mission science—advancing a predictive framework that connects the evolution of life on Earth to its potential forms beyond our planet.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.
Data accessibility. This article has no additional data.
Declaration of AI use. We have not used AI-assisted technologies in creating this article.

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Authors' contributions. A.S.: conceptualization, funding acquisition, project administration, supervision, writing—original draft, writing—review and editing; A.M.R.: validation, writing—original draft, writing—review and editing; M.M.: funding acquisition, writing—original draft, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

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